THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 4: **Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Burning of sugar (caramel, black) is: 1

(a) periodic **(b)** reversible
(c) a physical change **(d)** a chemical change

2. Which of the following can be used to show that a new substance has formed? 1

(a) change in size only **(b)** change in colour and gas formation
(c) change in shape only **(d)** change in position

3. A zinc-coated pipe is scratched. The iron below: 1

(a) melts **(b)** burns
(c) rusts at once **(d)** still remains protected

4. Heating copper sulphate crystals turns them white because: 1

(a) they melt **(b)** they freeze
(c) water of crystallisation is lost **(d)** they burn

5. The brown colour on a cut apple left in air is due to: 1

(a) melting **(b)** a physical change
(c) evaporation **(d)** a chemical change

6. Rusting is a: 1

(a) slow chemical change **(b)** reversible change
(c) periodic change **(d)** fast physical change

7. Adding water to the white powder turns it blue again. This is: 1

(a) physical change only **(b)** melting
(c) a reversible chemical change **(d)** burning

8. The change of milk to curd is: 1

(a) physical and reversible **(b)** periodic
(c) chemical and irreversible **(d)** reversible

9. A student claims that rusting is physical since iron is still iron. The correct reply is: 1

(a) iron changes shape only **(b)** iron melts
(c) iron dissolves **(d)** a new substance, iron oxide, forms

10. Which of the following changes can be reversed by heating or cooling? 1

(a) rusting **(b)** cooking
(c) burning **(d)** melting and freezing

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does sugar syrup cooled slowly give crystals but not on quick cooling? 2

12. Explain why a nail kept in oil does not rust. 2

13. Why is the process of milk getting curdled considered irreversible? 2

14. Explain why iron objects near the sea need extra protection. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why we cannot get back the paper from ash. 3

16. Which changes occur in a firework and how are they chemical? 3

17. How is a reversible chemical change different from a physical change? Give an example. 3

OR

List the observations during the burning of magnesium ribbon and give the nature of the change. 3

18. Explain why rusted iron weighs more than the original piece. 3

OR

Why is it risky to leave a wet iron bucket outside for long? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain with examples how a physical change can be turned into a chemical change by heating. 5

OR

Explain how nature maintains periodic changes like seasons and the Moon's phases. 5

20. Explain with a labelled description how to prepare a large crystal of alum. 5

OR

Describe the changes that occur in a pressure cooker while cooking pulses. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND